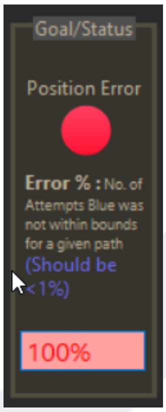
RL Problem Statement - Application Manual

Objective

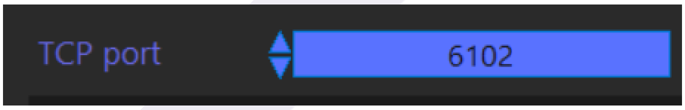
Control the Blue Dot by sending it position commands over TCP so that it tracks the Green Dot as closely as possible, minimizing the position error between the two in both X and Y.

Note: Open the application in **'administrator mode'** for access to loading and saving



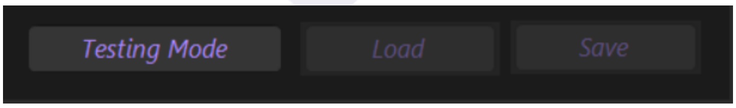
Connection

- The application listens on a configurable **TCP port** (default shown: 6102).
- All communication is over this single TCP connection: the application streams position/error data to you, and you send commands back to it.



Operating Modes

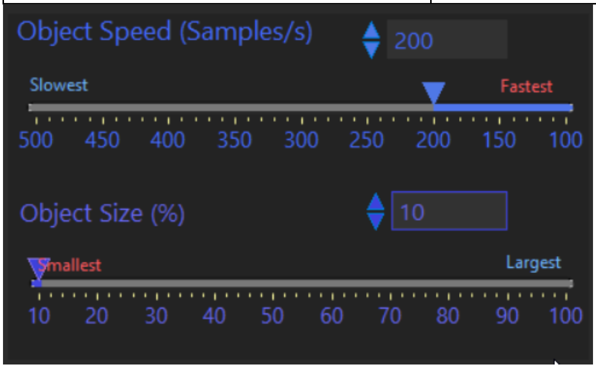
- **Testing Mode** — the Green Dot follows a new random path each run.
- **Learning Mode** — the Green Dot follows a fixed path.
    - **Save** stores the most recently traced path.
    - **Load** reloads a saved path so the Green Dot repeats the same path on loop.
- Load/Save are only active while in Learning Mode.



On-Screen Controls

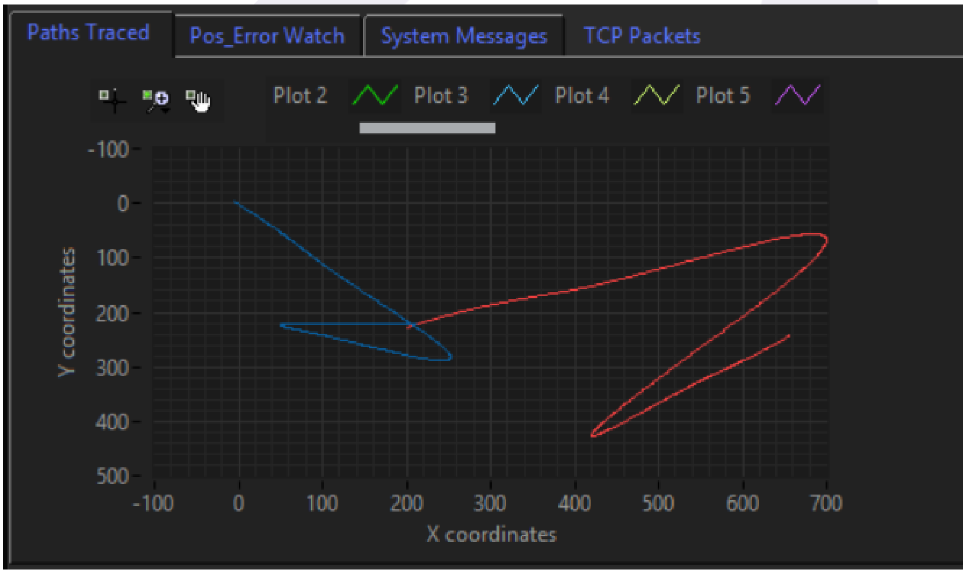
Control	Range	Notes
Object Speed (Samples/s)	100–500	"Slowest" at 500 → "Fastest" at 100

Control	Range	Notes
Object Size (%)	10–100	"Smallest" at 10 → "Largest" at 100



Status / Monitoring Panels

- **Goal/Status**: shows a *Position Error* indicator (colored dot) and *Error %* — the percentage of attempts where the Blue Dot was not within bounds of the green dot for a given path.
- **Paths Traced**: plot of recent path(s) traced, in X/Y coordinates.
- **Pos_Error Watch**: graph of X and Y position error over time.
- **System Messages**: log of application/system messages.
- **TCP Packets**: log of raw TCP packets sent and received.



TCP Data Sent BY the Application (20 bytes per packet)

Field	Type	Bytes
Green Dot Position — X	i32	4
Green Dot Position — Y	i32	4
Blue Dot Position — X	i32	4
Blue Dot Position — Y	i32	4
Error X	bool	1
Error Y	bool	1
Carriage Return (\r)	char	1
Line Feed (\n)	char	1
Total		**20 bytes**

TCP Data Sent TO the Application

Every command must begin with a single mode character: **C** (Configuration) or **P** (Position), telling the application what kind of data follows.

- **C mode**: adjust Object Speed and Object Size.
- **P mode**: move the Blue Dot by sending its X and Y position.

Configuration Mode (C) — 17 bytes total

Used to set Object Speed and Object Size.

Field	Type / Range	Bytes
Mode	"C"	1
Carriage Return	char	1
Line Feed	char	1
Object Speed	U32, 100–500	4
Object Size	Double, 10–100	8
Carriage Return	char	1
Line Feed	char	1

Position Mode (P) — 13 bytes total

Used to set the Blue Dot's position.

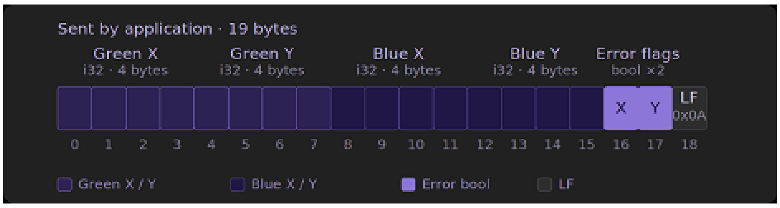
Field	Type	Bytes
Mode	"P"	1
Carriage Return	char	1

Field	Type	Bytes
Line Feed	char	1
Blue Dot Position — X	i32	4
Blue Dot Position — Y	i32	4
Carriage Return	char	1
Line Feed	char	1

TCP Data Sent by this Application : 20 Bytes

Sequence of Bytes:

Green Button Position : X (i32 4 bytes)
Green Button Position : Y (i32 4 bytes)
Blue Button Position : X (i32 4 bytes)
Blue Button position : Y (i32 4 bytes)
Error X : Bool (1 Byte)
Error Y : Bool (1 Byte)
Carriage Return Character : 1 Byte
Line Feed Character : 1 Byte



TCP data received by this Application

(To be sent by you to control the Blue dot or the configuration of the Application):

You need to send a char 'C' - Configuration or 'P' - Position, to denote what data you are sending to the Application, followed by the data.
*in **C mode**, you can adjust the Object speed & Object Size.*
*in **P mode**, you can move the Blue dot by sendinfg its x & Y position*

C Mode - To set Object Speed & Object Size : (17 Bytes)

Mode : "C" (1 byte)
Carriage Return Charactier : 1 Byte
Line Feed Character : 1 Byte

Object Speed (100-500) : (U32 4 bytes)
Object Size (10-100) : (Double 8 bytes)
Carriage Return Charactier : 1 Byte
Line Feed Character : 1 Byte

P Mode -To set the position of Blue Dot : (13 bytes)

Mode : "P" (1 byte)
Carriage Return Charactier : 1 Byte
Line Feed Character : 1 Byte

Blue Button Position : X (i32 4 bytes)
Blue Button position : Y (i32 4 bytes)
Carriage Return Charactier : 1 Byte
Line Feed Character : 1 Byte